

TADD MAO

Mathematics & Computer Science, University of Toronto

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RESEARCH INTERESTS

AI verification: how can we know, with formal certainty that an AI system did exactly what we asked, and how can we build a systems that can be trusted rather than trusting the model directly?

EDUCATION

University of Toronto

Sep. 2023 – Apr. 2027 (expected)

H.B.Sc., Computer Science Major (Focus in Artificial Intelligence) & Mathematics Major

Toronto, ON

- Cumulative GPA: **3.99 / 4.00**.
- Selected grades: Mathematical Logic (**100%**), Theory of Computation (99%), Abstract Mathematics (97%), Group Theory (97%), Real Analysis (94%), Advanced Calculus (94%), Machine Learning (90%).
- Additional coursework: Deep Learning, Artificial Intelligence, Natural Language Processing, Computational Linguistics, Topology, Graph Theory, Programming Languages.

PAPERS

Does the Proof Prove It That Way? Faithful Formalization of Euclid's Elements Proofs

Tadd Mao, Tianjun Zhong, Dhruva Arekar, Yuming Feng, One An, Jiani Huang, Xujie Si, Ziyang Li.

Preprint; under review at AAAI-27. [arXiv:2608.15432](#)

- Defined five necessary conditions for how faithfully a formal Lean proof reflects the natural-language proof it claims to represent, and built PISTIS, an oracle-guided proof search that generates proofs meeting them.
- On Euclid's *Elements* (Books I–III, 92 propositions): compiles at least 33× faster than prior work and is preferred 2.89× (127 blinded human reviews) and 5.2× (LLM judge) as often over the prior state of the art.
- Surfaced 52 gaps in Euclid's proofs and 2 genuine citation errors in the translated text.

Aligning Faithfulness for Theory-Scale Autoformalization

Yuming Feng, **Tadd Mao**, Dhruva Arekar, Tianjun Zhong, Jiani Huang, Xujie Si, Ziyang Li

Under review at ICLR 2027. [PDF](#)

- Found prior faithfulness evaluation methods ignore necessary context and implicit assumptions to properly evaluate faithfulness.
- Proposed a novel methodology to evaluate faithfulness to resolve this gap
- Built FAITHALIGN, the first theory-scale and context-aware faithfulness benchmark (460 items, 4,982 annotated phrase-level links), and found that across 8 models, the strongest reaches only 47.8% defect F1, establishing a large gap in LLM-as-judge.
- My contribution is that I first formalized how we label datapoints, created the UI that allows annotators to label the datapoints, and labelled the majority of the ProofNet section and all of the LeanEuclid/Pistis section of the dataset.

Which Reward Properties Allow Decentralized Value Learning to Scale in the Number of Agents?

Tadd Mao^{*}, Kirill Utyashev^{*}, Peter Marbach. ^{*}Equal contribution.

Under review at AAMAS 2027.

- Asked which *system* properties determine when decentralized value learning scales with respect to the number of agents while remaining accurate, and parametrized the reward system with two parameters; an agent's *interest* and *proficiency* to study this question.
- Found regions where decentralized scales well within our system, and regions where it does not, giving future work a way to identify when they should expect decentralized to scale.

RESEARCH GROUPS

Autoformalization & Faithfulness collaboration

May 2026 – present

Ziyang Li (Johns Hopkins) & Xujie Si (Toronto); with collaborators at JHU and U. Pennsylvania

Collaboration

- ↳ Building off of my faithful proof methodology (AAAI) and our novel faithfulness evaluation methodology + benchmark (ICLR), we are now working towards a faithfulness metric that performs well on our benchmark.

Autoformalization in Euclidean Geometry (NSERC USRA)

May 2026 – present

University of Toronto, advised by Prof. Xujie Si

Toronto, ON

- ↳ Also building off my AAAI submission where I formalized the first three books of Euclid's *Elements*, I am continuing to extend this to all of the books, which requires extending Avigad et al.'s SYSTEM E with a new axiom set (papers above).
- ↳ Developing methods for evaluating the quality of a given set of axioms.

OPEN-SOURCE CONTRIBUTION

- ↳ **Mathlib (Lean 4)** — authored and merged a lemma (`Int.closure_eq_zmultiples`) into MATHLIB, the community's core formal-mathematics library; wrote the proof independently and refined it through community review and design discussion. [PR #39435](#)

TEACHING

Teaching Assistant — MAT309 (Mathematical Logic)

Fall 2026

University of Toronto, with Prof. Soheil Homayouni

Toronto, ON

- ↳ Lead 2 of the 3 weekly tutorial sections; also marking and invigilation.

PROFESSIONAL SERVICE

- ↳ Sub-reviewer, AAAI 2027.

HONORS & AWARDS

- ↳ Nominee, CRA Outstanding Undergraduate Researcher Award (2026) — nominated by the Department of Computer Science at the University of Toronto.
- ↳ NSERC Undergraduate Student Research Award (2026).
- ↳ Dean's List Scholar (2024, 2025, 2026 — all completed years).
- ↳ Innis College Exceptional Achievement Award (2024).
- ↳ Later Life Learning Scholarship (2025).

SKILLS

Formal methods: Lean 4, Mathlib

Programming: Python, C/C++, Java, R, MATLAB

ML & tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas; Slurm, Linux, Git

Natural languages: English, French, Mandarin

PERSONAL

- ↳ Winner, Prince George Symphony Orchestra Young Artists Competition — performed the first movement of Rachmaninoff's Piano Concerto No. 2 as piano soloist with the PGSO ([video](#))
- ↳ Toured with professional musicians (Yu Yu Liu, Simon Macdonald, Roxanne Dykstra), performing violin in ensemble and piano solo, including a standing ovation for Liszt's *La Campanella*.
- ↳ Royal Conservatory of Music: ARCT diploma (piano); Level 10 (violin).

